

Portfolio Rebalancer User Guide

A tax-aware tool for keeping your investments on track

<https://frontend-sooty-three-ybjekuhhod.vercel.app>

1. What Is the Portfolio Rebalancer?

The Portfolio Rebalancer is a personal finance tool that helps you keep your investment portfolio aligned with your target asset allocation. It ingests your current holdings from a brokerage export, compares them against targets you set, and produces a concrete, tax-aware trade plan (which accounts to buy and sell in, and how much) to bring you back into balance.

All portfolio data stays **in your browser**. Nothing about your account names, dollar amounts, or positions is sent to a server unless you explicitly save a snapshot. The only things stored server-side are ticker classifications (shared) and encrypted PIN-protected snapshots (optional).

Key Capabilities

- Upload Schwab position exports (or a simple CSV template) for one or more accounts.
- Set your target allocation across 9 asset classes and see your current drift at a glance.
- Get an execution-ready buy/sell trade plan, tax-optimised per account.
- Project portfolio growth 5-40 years forward with Monte Carlo simulation.
- Compare against benchmark allocations (60/40, global equity, etc.).
- Identify tax-loss harvesting opportunities in taxable accounts.
- Get AI-generated tax-location advice (optional, requires Gemini API key).
- Save and reload encrypted snapshots across sessions.

2. Getting Started

2.1 Open the App

Open your browser and go to:

<https://frontend-sooty-three-ybjekuhhod.vercel.app>

***Note:** The backend service runs on Render's free tier and sleeps after 15 minutes of inactivity. The first request after a quiet period may take 30-60 seconds to wake up. A warm-up banner will appear while this happens.*

2.2 Export Your Holdings From Schwab

The app natively reads Charles Schwab position exports. To export:

- Log in to Schwab at schwab.com.
- Go to **Accounts** > choose the account > **Positions** tab.
- Click **Export** (top-right of the positions table) and save the .csv file.
- Repeat for each account you want to include (one file per account).

Alternatively, download the CSV template from the Setup page (click **Download template**) and fill it in manually with your holdings.

2.3 Upload and Set Targets

- **Upload CSV(s)** - click the Upload button and select one or more Schwab exports. Unknown tickers are classified automatically from their descriptions.
- **Review accounts** - each uploaded file appears as an account row. Adjust the account type (Taxable / Tax-Deferred IRA / Tax-Free Roth) if the auto-detected type is wrong.
- **Set target %** - enter your desired allocation % for each of the 9 asset classes. The total must equal 100%.
- **View Dashboard** - click the button to proceed to the analysis.

3. Understanding the Dashboard

3.1 Summary Strip

The top row of cards shows your total portfolio value, the number of accounts loaded, and your overall Asset Location Grade (1-10). Two action buttons sit in the top card:

- **Refresh prices** - fetches current market prices for all your tickers and updates current values without changing your uploaded data. Prices are cached for 24 hours.
- **Reset settings** - resets all sliders and controls to their defaults without losing your uploaded holdings or targets.

The Accounts card also lets you **add more accounts** (upload additional CSVs) or **remove an account** (click the X next to any account row) directly from the Dashboard, without going back to Setup.

3.2 Allocation Bars

The allocation bar chart shows each asset class as a horizontal bar. Blue = current %, grey = target %. A class marked *on target (within band)* is within your chosen drift tolerance and no trade will be generated for it.

3.3 Rebalancing Strategy

Two controls govern how the trade plan is generated:

- **Strategy slider (Minimize Gains)** - drag right to reduce the capital gains realized in taxable accounts. At 0 (left), the plan is purely optimal. At 100 (right), taxable sells are avoided as much as possible.
- **Drift band** - set a tolerance percentage (e.g. 3%). Asset classes within +/-3% of their target are left alone. Set to 0% to always rebalance to exact targets.

3.4 Trade Plan

The Proposed Trades table lists every buy, sell, or hold action, grouped by account. Each trade shows:

Column	Meaning
Account / Type	Which account the trade occurs in and its tax treatment.
Action	BUY (green), SELL (red), or HOLD (grey).

Asset Class	Which of the 9 sub-classes this trade addresses.
Ticker	The specific fund or security (if identifiable from your holdings).
Amount	Dollar amount to buy or sell.
Est. Gain	Estimated capital gain that would be realized (taxable sells only).
Tax note	Contextual note (e.g. 'no taxable event', 'may realize capital gains').

The plan is **cash-neutral per account**: each account's sells fund its own buys. No money moves between accounts. Use the **Export CSV** or **Print** buttons to save the plan for your records.

3.5 Projection Chart

The projection shows a Monte Carlo fan chart of possible portfolio growth over your chosen horizon (default 20 years). Key controls:

- **Horizon** - drag the slider to set the number of years to project (1-40).
- **Real vs Nominal** - toggle to show returns adjusted for inflation or raw dollar values.
- **Inflation %** - set your assumed inflation rate (default 2.5%).
- **Monthly contribution** - add or subtract a monthly amount (negative = withdrawal).
- **Net of fees** - check to subtract your portfolio's weighted expense ratio from returns.
- **Benchmark** - overlay a second projection line using a preset (60/40, 100% US, etc.) or a custom stock/bond split. The benchmark always projects at 0% fees for fair comparison.

The fan shows the 10th, 25th, 50th (median), 75th, and 90th percentile outcomes across 1,000 simulated paths. A deterministic (steady-return) line is also shown.

3.6 Tax-Loss Harvesting Panel

If any of your **taxable** holdings have an unrealized loss (and a known cost basis), they appear here sorted by loss size. Selling to harvest the loss can offset capital gains or income.

***Wash-sale rule:** do not repurchase the same or substantially identical fund within 30 days before or after the sale, or the loss will be disallowed.*

3.7 Asset Location Grade

The grade (1-10) measures how well your current holdings are placed across account types according to tax-efficiency principles. A score of 10 means every asset class is in the ideal account type. Common reasons for a lower score:

- Tax-inefficient bonds or REITs sitting in a taxable account.
- Tax-efficient equity (broad index funds) in a tax-deferred account instead of Roth/taxable.
- High-growth assets (crypto, small-cap) not in tax-free (Roth) accounts.

3.8 AI Tax-Location Advisor

Click **Get insights** to request Gemini AI analysis of your specific allocation. The advisor looks at your grade breakdown and per-account holdings and returns 3-5 actionable suggestions. Results are cached so

repeated requests for the same portfolio state do not consume additional API quota.

3.9 Scenario / What-If Panel

The Scenario panel lets you edit a hypothetical copy of your holdings (change amounts, add/remove positions) and see the rebalancing result side-by-side with your actual portfolio, without changing your real data.

4. Saving and Loading Snapshots

Snapshots let you save a point-in-time capture of your portfolio and reload it later, even after closing the browser. Navigate to the **Snapshots** page from the top bar.

Privacy Note

Snapshots contain only tickers, quantities, and account TYPES (taxable/IRA/Roth).
Account names and dollar amounts are NOT stored server-side.
The payload is encrypted with a server-side Fernet key before being saved to the database.
You can also download a local JSON copy that stays entirely on your machine.

To save a snapshot, enter a label and your chosen PIN, then click Save. To reload, enter your email and PIN on the Load tab and select from your saved snapshots.

5. The 9 Asset Sub-Classes

The app tracks 9 sub-classes. Bonds and Alternatives are split by tax efficiency so the rebalancer can route them to the right account type.

Sub-Class	Description & Tax Profile
US Stock	Broad domestic equity index funds (e.g. VTI, VOO, SCHB). Tax-efficient.
International	Developed and emerging markets ex-US (e.g. VXUS, VEA, VWO). Tax-efficient.
Muni Bond	Municipal bonds - interest is federally tax-exempt. Tax-efficient; ideal in taxable.
Taxable Bond	Corporate and government bonds (e.g. BND, AGG, TIP). Tax-inefficient; ideal in tax-deferred.
REITs	Real estate investment trusts. High-dividend; tax-inefficient; ideal in tax-deferred.
Cash	Money-market funds, short-term T-bills, HYSA. Tax-neutral.

Gold & Commodities	Gold ETFs, commodity funds (e.g. GLD, IAU, PDBC). Tax-inefficient.
Crypto	Bitcoin ETFs, crypto funds (e.g. IBIT, FBTC). High volatility; tax-inefficient.
Other Alternatives	Private equity, hedge funds, timber, infrastructure. Tax-inefficient.

6. CSV Upload Format

The app accepts two CSV formats:

Format A: Schwab Position Export

Download directly from schwab.com > Positions > Export. The account name and type are extracted automatically from the header row.

Format B: Simple Template

A flat CSV with these columns (download the template from the Setup page):

Column	Notes
account_name	Any label, e.g. 'Brokerage' or 'Roth IRA'
account_type	taxable, tax_deferred, or tax_free
ticker	Fund or stock symbol, e.g. VTI
quantity	Number of shares held
cost_basis	Total cost basis in dollars (0 if unknown)
current_value	Current market value in dollars

Multiple accounts can coexist in a single template file (just use different account_name values per row) or be split across multiple files.

7. Tips for Best Results

- **Include all your accounts** - the rebalancer routes assets across accounts for optimal placement, so the more accounts you include, the better the location decisions.
- **Set cost basis where you know it** - the tax-loss harvesting panel only shows holdings with a known cost basis. Leave as 0 if unknown.
- **Refresh prices before rebalancing** - click Refresh Prices so the trade sizes are based on current values, not your last export.

- **Use the drift band for low-turnover rebalancing** - a 3-5% band keeps you close to your targets without triggering trades for minor drifts.

- **The trade plan is a recommendation, not an instruction** - always verify amounts in your brokerage before placing trades, especially for fractional shares.